Cell Tuition
1-1 PRIVATE TUTORS

MEASUREMENT

Volume of Prisms

Name _____ Date _____ Calculator allowed Version 1

Scan to practise this topic online

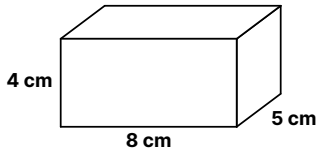
KEY WORDS prism *(poro)* · cross-section *(hautanga whakapae)* · cylinder *(porotaka)* · capacity *(rōrahi)*

A Cuboids ●○○

Remember: Volume = length × width × height. The answer is in cubic units: cm^3, m^3 or mm^3.

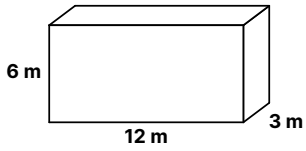
1. Find the volume of each solid.

a)



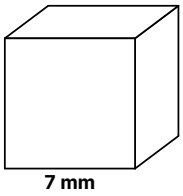
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b)



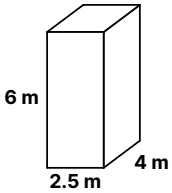
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c)



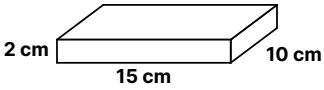
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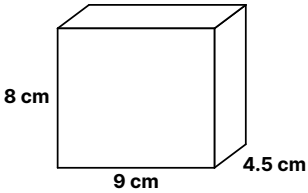
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e)



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f)



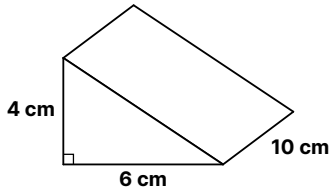
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B Triangular and trapezium prisms ●●○

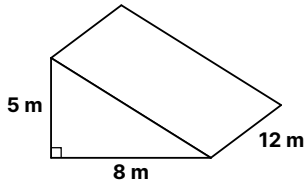
Remember: Volume of any prism = area of the cross-section × length. Find the area of the end first.

2. Find the volume of each prism.

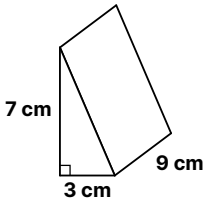
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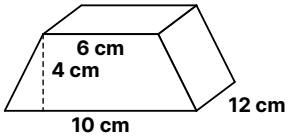
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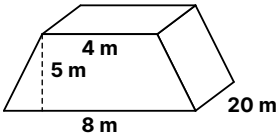
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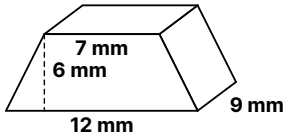
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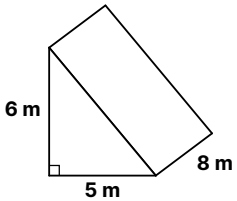
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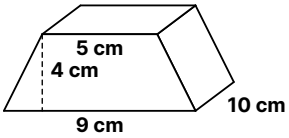
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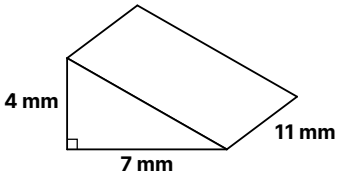
g)



h)



i)



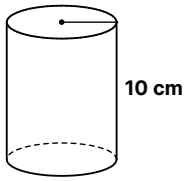
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C Cylinders ●●○

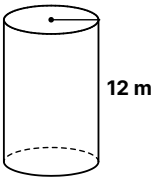
Remember: A cylinder is a prism with a circular cross-section: Volume $= \pi r^2 h$. Halve a diameter to get the radius first. Give answers to 1 decimal place.

3. Find the volume of each cylinder.

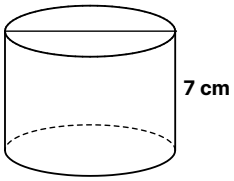
a)



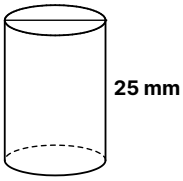
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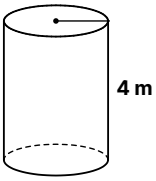
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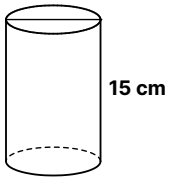
d)



e)



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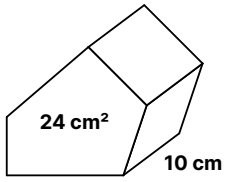


D When the cross-section is given ●●○

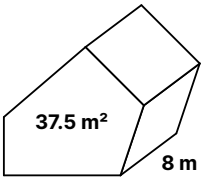
Remember: If the area of the end is given, just multiply it by the length. Check that the area and the length use the same unit.

4. Find the volume of each prism.

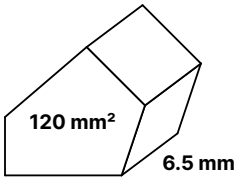
a)



b)



c)



E Missing lengths and litres ●●●

Remember: Put what you know into the formula and solve for the missing length. 1 litre = 1000 cm³, and 1 m³ = 1000 litres.

5. Find the missing length or change the units.

a) A rectangular prism has a volume of 240 cm³ and a base 8 cm by 5 cm. Find its height.

b) A prism has a volume of 450 m³ and a cross-section of area 25 m². Find its length.

c) A cylinder has a volume of 600 cm³ and a radius of 4 cm. Find its height, to 1 decimal place.

d) A rectangular prism has a volume of 126 mm³ and a base 7 mm by 3 mm. Find its height.

e) Change 4,500 cm³ to litres. (1 L = 1000 cm³)

f) Change 2.4 m³ to litres. (1 m³ = 1000 L)

g) A cylinder has a volume of 1,000 cm³ and a radius of 6 cm. Find its height, to 1 decimal place.

h) Change 850 cm³ to litres. (1 L = 1000 cm³)

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F Problems ●●●

6. A gold bar is a prism whose end is a trapezium with parallel sides of 5 cm and 7 cm, 3 cm apart. The bar is 20 cm long. Find its volume.

7. A circular pond has a radius of 2 m and is 0.5 m deep all over. How many litres of water does it hold, to the nearest litre?

8. A raised garden bed is 2.4 m by 1.2 m and 0.5 m deep. Soil costs $65 per cubic metre. Find the cost of filling the bed.

9. A swimming pool is 25 m long and 10 m wide. It is 1.2 m deep at the shallow end and 2.4 m deep at the deep end, with the floor sloping evenly. How many litres does it hold?

10. A cylindrical water tank has a radius of 40 cm and a height of 90 cm. How many litres does it hold, to the nearest litre?

11. A concrete path is 18 m long, 1.2 m wide and 0.1 m thick. Concrete costs $185 per cubic metre. Find the cost of the concrete.

12. A tent is a triangular prism. Its end is a triangle with a base of 2.4 m and a height of 1.8 m, and the tent is 3 m long. Find the volume of air inside.

★ Extension

13. A cylinder has radius 4 cm and height 10 cm. Which increases the volume more: doubling the radius, or doubling the height? Explain with the formula.

14. A rectangular prism has a width of x cm, a length of $2x$ cm and a height of $(x + 3)$ cm. Write an expression for its volume, expanded.

15. A metal pipe has an outer radius of 6 cm, an inner radius of 5 cm and is 200 cm long. Find the volume of metal in the pipe, to the nearest cm³.

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Volume of Prisms

Name _____ Date _____ Calculator allowed Version 2

Scan to practise this topic online

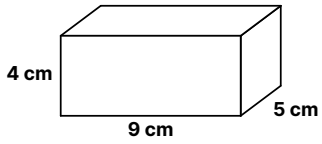
KEY WORDS prism *(poro)* · cross-section *(hautanga whakapae)* · cylinder *(porotaka)* · capacity *(rōrahi)*

A Cuboids ●○○

Remember: Volume = length × width × height. The answer is in cubic units: cm^3, m^3 or mm^3.

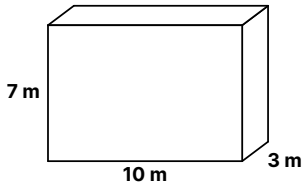
1. Find the volume of each solid.

a)



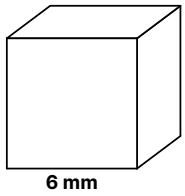
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b)



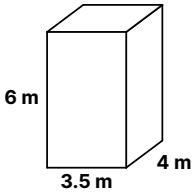
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c)



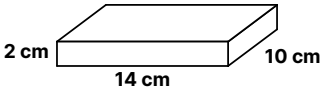
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d)



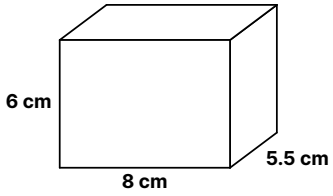
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e)



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f)



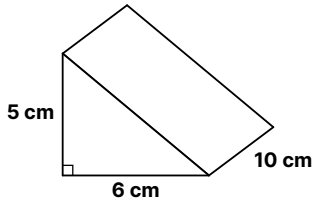
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B Triangular and trapezium prisms ●●○

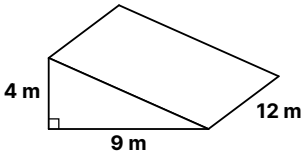
Remember: Volume of any prism = area of the cross-section × length. Find the area of the end first.

2. Find the volume of each prism.

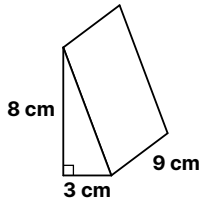
a)



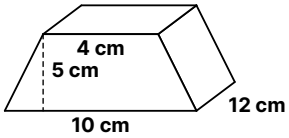
b)



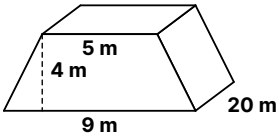
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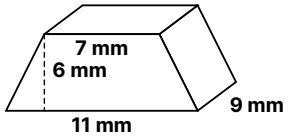
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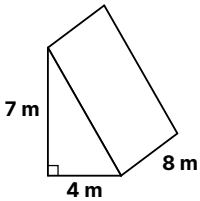
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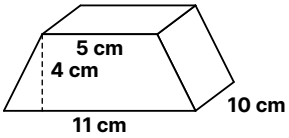
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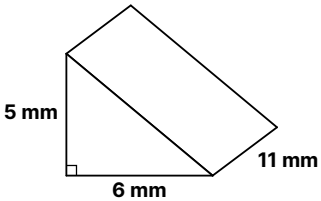
g)



h)



i)



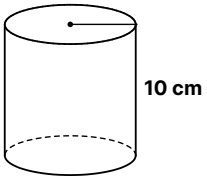
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C Cylinders ●●○

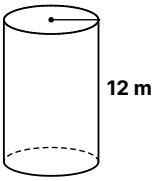
Remember: A cylinder is a prism with a circular cross-section: Volume $= \pi r^2 h$. Halve a diameter to get the radius first. Give answers to 1 decimal place.

3. Find the volume of each cylinder.

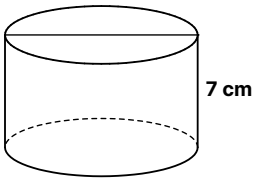
a)



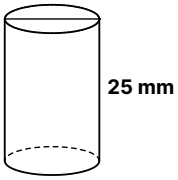
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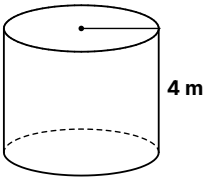
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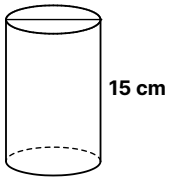
d)



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f)

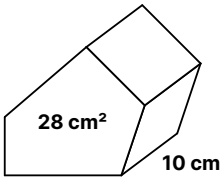


D When the cross-section is given ●●○

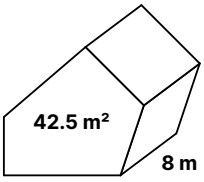
Remember: If the area of the end is given, just multiply it by the length. Check that the area and the length use the same unit.

4. Find the volume of each prism.

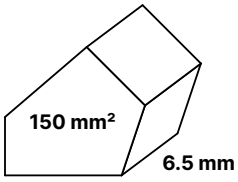
a)



b)



c)



E Missing lengths and litres ●●●

Remember: Put what you know into the formula and solve for the missing length. 1 litre = 1000 cm³, and 1 m³ = 1000 litres.

5. Find the missing length or change the units.

a) A rectangular prism has a volume of 280 cm³ and a base 7 cm by 5 cm. Find its height.

b) A prism has a volume of 540 m³ and a cross-section of area 30 m². Find its length.

c) A cylinder has a volume of 800 cm³ and a radius of 5 cm. Find its height, to 1 decimal place.

d) A rectangular prism has a volume of 144 mm³ and a base 8 mm by 3 mm. Find its height.

e) Change 6,500 cm³ to litres. (1 L = 1000 cm³)

f) Change 3.2 m³ to litres. (1 m³ = 1000 L)

g) A cylinder has a volume of 1,200 cm³ and a radius of 7 cm. Find its height, to 1 decimal place.

h) Change 1,250 cm³ to litres. (1 L = 1000 cm³)

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F Problems ●●●

6. A gold bar is a prism whose end is a trapezium with parallel sides of 6 cm and 8 cm, 3.5 cm apart. The bar is 24 cm long. Find its volume.

7. A circular pond has a radius of 1.5 m and is 0.6 m deep all over. How many litres of water does it hold, to the nearest litre?

8. A raised garden bed is 3 m by 1.5 m and 0.4 m deep. Soil costs $70 per cubic metre. Find the cost of filling the bed.

9. A swimming pool is 20 m long and 8 m wide. It is 1 m deep at the shallow end and 2.6 m deep at the deep end, with the floor sloping evenly. How many litres does it hold?

10. A cylindrical water tank has a radius of 35 cm and a height of 100 cm. How many litres does it hold, to the nearest litre?

11. A concrete path is 24 m long, 1.5 m wide and 0.1 m thick. Concrete costs $190 per cubic metre. Find the cost of the concrete.

12. A tent is a triangular prism. Its end is a triangle with a base of 2.6 m and a height of 2 m, and the tent is 3.5 m long. Find the volume of air inside.

★ Extension

13. A cylinder has radius 3 cm and height 12 cm. Which increases the volume more: doubling the radius, or doubling the height? Explain with the formula.

14. A rectangular prism has a width of x cm, a length of $2x$ cm and a height of $(x + 3)$ cm. Write an expression for its volume, expanded.

15. A metal pipe has an outer radius of 8 cm, an inner radius of 7 cm and is 150 cm long. Find the volume of metal in the pipe, to the nearest cm³.

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